FORMALIZING SMALL-ENERGY REGULARITY AND STRATIFICATION OF MINIMIZING HARMONIC MAPS IN LEAN

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Abstract. We describe a Lean 4 formalization of interior small-energy regularity and classical singular-set stratification for energy-minimizing harmonic maps. The regularity development separates the passage from small energy to a Hölder representative from the analytic upgrade to smoothness. The second development imports this interface and proves Luckhaus interpolation, strong compactness, tangent-map structure and the Hausdorff dimension bounds for singular strata. We give linked dependency diagrams and mathematical statements corresponding to the public Lean declarations. We also examine the constants hidden by existential packaging and extract the explicit arithmetic of the derivative bootstrap. This yields a parameterized recurrence, while identifying the analytic and geometric inputs still needed for numerical evaluation. The account emphasizes proof selection, shared witnesses and dimension-independent induction. The implementation used Codex with Scott Armstrong's Lean skills, with mathematical supervision at the level of statements and proof architecture.

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1. Introduction

Formalizing a regularity theorem requires more than translating its last estimate. The functions in a comparison argument must belong to the right Sobolev spaces. Their traces must agree, their weak derivatives must be identified, and every application of minimality must use an admissible competitor. These obligations become particularly visible when regularity is used inside a compactness and dimension-reduction argument.

This article reports two linked developments. The first proves smooth small-energy regularity for minimizing harmonic maps [10]. The second proves the classical stratification theorem, including

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the isolation of interior singularities in dimension three [11]. Their mathematical source is the presentation in Giaquinta–Martinazzi [4]. The underlying regularity theorem is due to Schoen and Uhlenbeck [7]. The compactness argument uses Luckhaus interpolation [6].

The formalization was organized around a mathematical choice made before large-scale proof generation: which complete proof should be supplied to the agent? Different proofs of the same regularity theorem make very different demands on a formal library. A direct estimate can expose constant dependence. A compactness argument can be shorter on paper but require substantial infrastructure before it produces a contradiction. Neither route is uniformly preferable.

1.1. **Conventions and main results.** Let $n \geq 2$ and let $N \subset \mathbb{R}^q$ be a nonempty compact smooth embedded submanifold without boundary, with the induced metric. Let $\Omega \subset \mathbb{R}^n$ be open. A minimizing map in this article is an N-valued map $u \in W^{1,2}_{\mathrm{loc}}(\Omega, \mathbb{R}^q)$ that minimizes energy on every ball compactly contained in Ω, against all N-valued Sobolev competitors whose difference from u belongs to $W^{1,2}_0$ on that ball. The Lean predicate retains the displayed weak gradient Du. We use

$$E(u; B) = \int_B |Du|^2_{\mathrm{HS}} \, \mathrm{d}x, \qquad \Theta_u(a, r) = r^{2-n} E(u; B_r(a)). \tag{1.1}$$

There is no factor $1/2$ in the energy. Higher derivatives use ambient multilinear operator norms. A regular point is a point with a neighborhood on which u admits a smooth almost-everywhere representative. The singular set $\Sigma(u)$ is its complement in Ω.

Theorem 1.1 (Smooth small-energy regularity)**.** *For each n, N there exist $\varepsilon_* > 0$, $0 < \gamma < 1$, and a sequence of finite nonnegative constants $(C_k)_{k \geq 0}$ such that the following holds. If u is minimizing, $\overline{B_R(a)} \subset \Omega$ and $\Theta_u(a, R) \leq \varepsilon_*$, then u has a unique continuous representative w on $B_{R/2}(a)$. It is smooth, takes values in N and satisfies*

$$\sup_{B_{R/2}(a)} |w - w(a)| \leq C_0 \sqrt{\Theta_u(a, R)}, \tag{1.2}$$

$$R^k \sup_{B_{R/2}(a)} \|D^k w\| \leq C_k \sqrt{\Theta_u(a, R)} \qquad (k \geq 1), \tag{1.3}$$

$$R^{k+\gamma} [D^k w]_{\gamma; B_{R/2}(a)} \leq C_k \sqrt{\Theta_u(a, R)} \qquad (k \geq 1). \tag{1.4}$$

The threshold, exponent and whole sequence are chosen before the domain, map, weak gradient, ball and derivative order. All conclusions concern the same representative. Uniqueness is on $B_{R/2}(a)$.

The exact declaration is `HolderToCK.smallEnergySmoothRegularity`. It includes dimension two and zero energy. It does not assert a prescribed ambient extension of w outside the half ball.

Write $\Theta_u(a) = \lim_{r \downarrow 0} \Theta_u(a, r)$. For a tangent map φ, let $S(\varphi)$ be its spine, whose carrier is $\{z : \Theta_\varphi(z) = \Theta_\varphi(0)\}$. These objects and their qualifications are stated in Appendix A. Define

$$S_j(u) = \{a \in \Sigma(u) : \dim S(\varphi) \leq j \text{ for every tangent map } \varphi \text{ at } a\}. \tag{1.5}$$

In particular, the quantifier is universal over actual tangent limits.

Theorem 1.2 (Interior stratification)**.** *Let u be minimizing. If $n \geq 3$, then*

$$S_0(u) \subset \cdots \subset S_{n-3}(u) = S_{n-2}(u) = S_{n-1}(u) = \Sigma(u).$$

For every $0 \leq j \leq n-3$ and every real $s > j$,

$$\dim_{\mathrm{H}} S_j(u) \leq j, \qquad \mathcal{H}^s(S_j(u)) = 0.$$

Consequently $\dim_{\mathrm{H}} \Sigma(u) \leq n-3$ and $\mathcal{H}^s(\Sigma(u)) = 0$ for every $s > n-3$. If $n = 3$, the singular set has no accumulation point anywhere inside Ω. If $n = 2$, it is empty.

The higher-dimensional statement is `StratificationTheory.stratification`. Its density-concentration clause is given in Proposition A.13. The two-dimensional conclusion is the separate export `singularSet_eq_empty_two_dimensional`. Interior isolation permits accumulation at the boundary and does not imply that the full singular set is finite. The development does not include boundary regularity, rectifiability or quantitative stratification.

1.2. **Relation to other formalization projects.** Armstrong and Kempe's formalization of De Giorgi–Nash–Moser theory [1] provides a useful model for presenting analysis in Lean: state the mathematical results, explain the analytic organization, and describe what was actually checked. Their Moser iteration also motivates examining the constants in an iterative proof rather than leaving every occurrence of C uninterpreted. The present project uses Armstrong's *LeanAutoformalizationSkills* [3]. The skills supply instructions for statement audits, dependency planning, proof development and verification.

Related recent work includes the Lean formalization of the Caffarelli–Kohn–Nirenberg theorem [2] and Zhang et al.'s Poincaré-conjecture project [12]. These comparisons concern methods of organizing substantial formal developments. The results and trust boundaries of each project must be read from its own statements.

2. Selecting and separating the proof

2.1. **Why this regularity argument.** Giaquinta–Martinazzi, Section 10.3.1, presents the original Schoen–Uhlenbeck route and explicitly distinguishes it from a route based on a Caccioppoli inequality. The latter uses Luckhaus interpolation and harmonic approximation, as in Simon's treatment [8]. Both are valid approaches. A third useful viewpoint is the compactness and contradiction strategy associated with the regularity theory of Hardt and Lin [5]. Appendix B describes these alternatives and their different quantitative commitments.

The chosen argument uses uniform smoothing, harmonic replacement, projection to a tubular neighborhood, and an energy comparison. Its advantage for this project was the visibility of the comparison scales and smallness conditions. It could be developed before the full Luckhaus gluing machinery. This mattered because the regularity theorem was needed as an input to the later compactness theory.

This choice does not eliminate harmonic functions or all uses of compactness. Harmonic replacement is part of the proof. Compactness of the target supplies uniform geometric bounds. Moreover, the current Lean proofs select some analytic constants existentially. The useful distinction is between a direct chain of estimates and an argument whose decay threshold is obtained only by excluding a sequence of counterexamples. An explicit numerical threshold requires quantifying every input in either chain.

2.2. **Small energy to Hölder continuity.** Local minimality first gives inner stationarity and the monotonicity formula. Outer variation gives the weak Euler equation. Fixed smoothing produces an ambient smooth map close to N. Harmonic replacement controls the energy of its projection. Variable smoothing joins this map to u across a shell, with the original Sobolev boundary values. The resulting competitor is admissible in the definition of local minimality.

The comparison proves one-step decay, stated in Lemma A.3. Rescaling and iteration then give, at every admissible center and every sufficiently small scale,

$$\Theta_u(x,r) \le 2(r/s)^{2\alpha}\Theta_u(x,s), \qquad \alpha = \frac{\log 2}{2\log(1/\tau)}, \quad 0 < r \le s. \tag{2.1}$$

Here $\tau^{2\alpha} = 1/2$. A change of center and a Morrey estimate produce a continuous Hölder representative, including almost-everywhere agreement, pointwise target values and local uniqueness. This is Proposition A.4.

The formal argument retains details easily suppressed in a paper proof. The variable-radius smoothing derivative contains the full matrix $I - y \otimes Ds$ and the Jacobian $1 - Ds \cdot y$. The comparison map must have a difference in the actual function-and-gradient closure defining $W_0^{1,2}$. Establishing these facts is part of the proof, rather than part of the definition of a minimizer.

2.3. **The independent analytic upgrade.** Once a Hölder representative is available, its smoothness is an elliptic regularity problem. A smooth coefficient A obtained from the target projection gives the signed equation

$$\Delta w = F, \qquad F = \sum_{i=1}^{n} A(w)[D_i w, D_i w]. \tag{2.2}$$

The upgrade library proves a generic quadratic-Poisson regularity theorem before applying it to harmonic maps. The shared interfaces are the map, its displayed weak gradient, the target and the signed weak equation.

The first step is genuinely analytic. If h is the harmonic replacement on B_r, $[w]_\alpha \le H$, and $|F| \le K|Dw|_{\mathrm{HS}}^2$, the proved comparison gives

$$\int_{B_r} |D(w-h)|_{\mathrm{HS}}^2 \le 2KHr^\alpha \int_{B_r} |Dw|_{\mathrm{HS}}^2. \tag{2.3}$$

Improved energy growth and gradient-excess decay yield a Campanato representative of the weak gradient. Mollification identifies it with the classical derivative of the same w. The resulting exponent is $\gamma = \alpha/8$. This gives the $C^{1,\gamma}$ input to the bootstrap.

The library also proves the weak Poisson Schauder estimate from the fundamental solution. Its Hessian cancellation includes the distributional point term with coefficient δ_{ij}/n and the correct sign. The differentiated equations and product estimates then give every higher derivative. The energy threshold is chosen once. The argument does not shrink it anew for each k.

Separating these tasks improves reuse and review. The Hölder theorem can be applied without importing the full higher-order analysis. The quadratic-Poisson theorem can be reused with coefficients not arising from harmonic maps. Joining the two theorems requires almost-everywhere uniqueness, so that the continuous and smooth representatives agree.

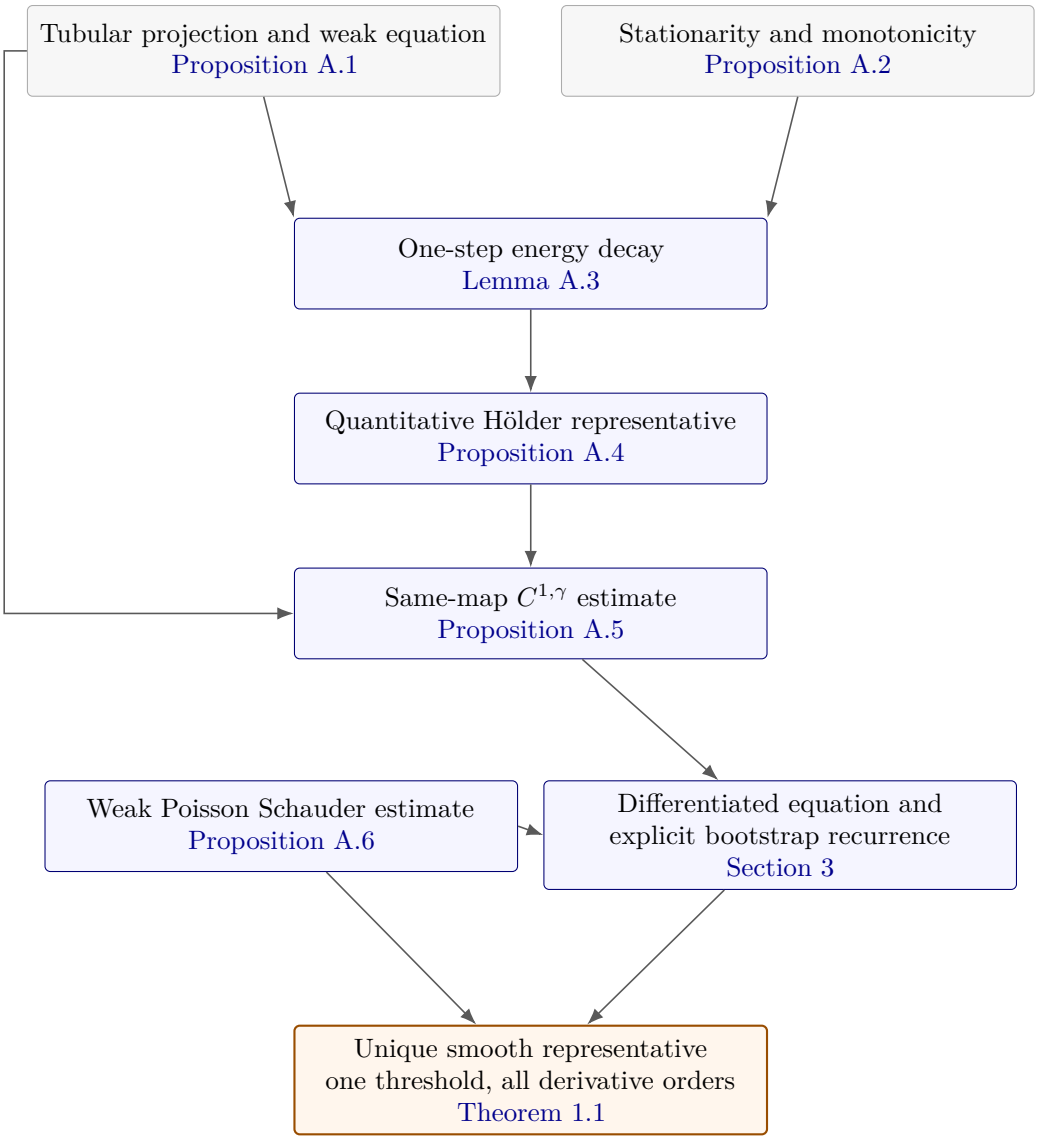


FIGURE 1. The regularity argument. Each numbered box links to its mathematical statement. The coefficient equation also enters the C^1 comparison. The Schauder estimate is a separate analytic input to the higher-order induction. This is a mathematical dependency diagram, not a claim that every import edge is shown.

3. Constants and existential choices

3.1. **What can be extracted from `Classical.choose`.** The occurrence of `Classical.choose` does not by itself determine whether a constant can be made explicit. In `UniformThreshold.lean`, the next derivative bound is selected from an existence theorem and then iterated by `Nat.rec`. Following that existence theorem into `NestedBalls.lean` reveals an explicit witness. The composition estimates and nested-radius Schauder bounds also have explicit arithmetic bodies.

Other choices need more work. The proof chooses small scales using convergence of powers at zero. Those choices can be replaced by elementary power formulas. Bounds on derivatives of a fixed smoothing kernel are obtained using compactness, but the kernel is an explicit function whose derivatives can be bounded directly. In contrast, an abstract smooth embedded target supplies no numerical tube radius or curvature bounds. These require quantitative geometric data for a particular target.

There are therefore three distinct statements. The existing Lean proof checks the existential theorem and its local witness estimates. The formulas below recombine those estimates into transparent admissible choices. The additional scalar inequalities used to eliminate some compactness choices are proved mathematically in Appendix C, but have not been added to Lean. The numerical evaluations are approximations of these formulas. They are not evaluations of the opaque witness selected by the public theorem.

3.2. **An explicit sufficient energy threshold.** Let $v_n = |B_1^n|$ and write s for the standard smooth transition,

$$s(t) = \frac{p(t)}{p(t) + p(1-t)}, \qquad p(t) = \begin{cases} e^{-1/t}, & t > 0, \\ 0, & t \le 0. \end{cases}$$

The fixed kernel is $s(1 - 4|x|^2)/I_n$, where

$$I_n = \int_{\mathbb{R}^n} s(1 - 4|x|^2)\,\mathrm{d}x, \qquad S_n = \frac{2^{2n-1}}{I_n}.$$

The smoothing proof exhibits this S_n. If $\rho > 0$ is a valid common projection radius and $K_N \ge \sup_N \|D^2\pi\|$, set

$$\begin{aligned} H_{n,N} &= 2\sqrt{S_n}(1 + K_N) + 4^n/v_n + 1, \\ B_{n,N} &= 8(1 + v_n)(1 + 2^n)H_{n,N} + 50. \end{aligned} \tag{3.1}$$

These are the actual scalar witnesses in the harmonic replacement and energy comparison proofs, after making coherent choices of the same smoothing constant. A sufficient replacement for the limit-selected threshold is

$$\boxed{\varepsilon_{\mathrm{dec}} = \min\left\{\frac{\rho^2}{4S_n},\ 16^{-64(n-1)},\ (6B_{n,N})^{-32(n-1)}\right\}.} \tag{3.2}$$

For $\beta = 1/[64(n-1)]$ the comparison bounds its error by $3B_{n,N}\eta^{2\beta}E(u; B_1)$. The last term in (3.2) makes this at most $E(u; B_1)/2$. For $0 < \eta \le \varepsilon_{\mathrm{dec}}$, put $h = \eta^{1/4}$, $d = \eta^{1/8}$ and $t = \eta^{\beta}$. The other terms give

$$\sqrt{S_n\eta} \le \rho/2, \qquad t \le 1/16, \qquad d \le 16^{-8}, \qquad d/t \le 16^{9-8n} < 1/6.$$

These imply the geometric conditions for the shell and projection construction. The fixed decay radius is $\tau = \varepsilon_{\mathrm{dec}}^{\beta}$. The source choices appear in `DecayParameters.lean` and `DecayThreshold.lean`.

No quadrature is necessary for a conservative value: monotonicity of s gives $I_n \ge v_n/(2\,4^n)$, hence $S_n \le 16^n/v_n$. This upper bound may replace S_n throughout the estimates. In dimension two, symmetry gives the sharper exact evaluation

$$I_2 = \pi/8, \qquad S_2 = 64/\pi,$$

so that

$$B_{2,N} = 40(1+\pi)\left[\frac{16}{\sqrt{\pi}}(1+K_N)+\frac{16}{\pi}+1\right]+50.$$

The dimensional numbers here are explicit. The target parameters are still required. A numerical bound on the second fundamental form alone should not be silently identified with the full ambient Hessian bound K_N.

The Hölder proof then gives

$$\varepsilon_H = 2^{2-n}\varepsilon_{\mathrm{dec}}, \qquad \alpha = \frac{\log 2}{2\log(1/\tau)}. \tag{3.3}$$

Its explicit Morrey coefficient and the initial C^1 constants are listed in Appendix C.

3.3. **A concrete recurrence for every derivative order.** Choose bounds $b_j \geq 0$ for $\|D^j A\|$ on a fixed closed ball containing the range of the map. Let M_1 be the initial bound for oscillation, gradient and its γ-Hölder seminorm after factoring out $\delta = \sqrt{E(u;B_1)} \leq 1$. For $m \geq 1$ use

$$r_m = \tfrac{1}{2} + 2^{-m-2}, \qquad \Delta_m = 2^{-m-3}, \qquad \rho_m = \Delta_m/2.$$

Let S_m be the explicit nested Schauder coefficient, and let P_{m-1}, Q_{m-1} be the explicit bounds for the differentiated quadratic forcing and its Hölder seminorm. With the arguments specified in (C.7) below, an admissible recurrence is

$$\boxed{M_{m+1} = M_m + n^{m+1}S_m(M_m + P_{m-1} + Q_{m-1}), \qquad C_0 = C_1 = M_1, \quad C_k = M_k \ (k \geq 1).} \tag{3.4}$$

This is extracted from the witness in the nested-step proof. It need not be definitionally equal to the sequence selected by `Classical.choose`.

The first nontrivial step is already informative. Put $d = (5/4)^{1-\gamma}$. Then

$$C_2 = M_1 + n^2 S_1\left[M_1 + n(b_0+b_1)(M_1 d)^2(M_1 d + 3)\right] \tag{3.5}$$

is an admissible choice. This displays the coefficient dependence rather than concealing it behind an unspecified constant. The full recurrence is arithmetically evaluable once the geometric and initial data are fixed.

3.4. **A numerical sphere specialization.** A complete mathematical specialization is possible for the unit sphere. Take $N = S^{q-1}$ with $q \geq 2$, $\rho = 1/8$ and $K_N = 6$. The nearest projection is $\pi(z) = z/|z|$, its derivative has norm at most $4/3$ on the tube $\mathrm{dist}(z,N) < 1/4$, and its ambient Hessian on N has norm at most 6. A smooth cutoff supplies a global extension, as described in Appendix C.

For the later elliptic bootstrap use $A(p)[v,w] = -p\langle v,w\rangle$. The sphere equation is $\Delta u = -u|Du|^2_{\mathrm{HS}}$. On the closed unit ball this coefficient has $b_0 = b_1 = 1$ and $b_j = 0$ for $j \geq 2$, and the forcing bound uses $K = 1$. Here $K = 1$ concerns the equation on tangent gradients. It is not a bound of 1 for the full ambient Hessian of nearest projection.

For $n = 3$, using the conservative $S_n \leq 16^n/v_n$ and the explicit universal bounds in the appendix gives the approximate values

$$\alpha \simeq 0.0501065, \qquad \gamma \simeq 0.00626332, \qquad \varepsilon_{\mathrm{dec}} \simeq 3.16926 \times 10^{-385}.$$

The common smoothness threshold and derivative coefficients are enormous:

Quantity	Base-ten logarithm of the extracted expression
ε_*	-2757.66039
$C_0 = C_1$	709.53037
C_2	2147.89570
C_3	8612.47861

These are deliberately crude bounds, not optimized estimates. The table reports approximate evaluations of exact expressions, not certified rounded numerical upper bounds. The geometric specialization and scalar replacements are an additional mathematical calculation, not an explicit-constant sphere theorem currently exported by the Lean project. The accompanying standalone scripts reproduce both this calculation and conditional higher-order recurrences without changing project code.

This example shows what can be extracted by inspecting the proof, while also showing why explicitness and useful sharpness are separate aims. For an unspecified compact smooth target, numerical values still require quantitative geometry. Formalization makes the missing inputs visible. It does not manufacture them from an abstract compactness proof.

4. From regularity to stratification

The second project imports the first at a fixed commit. It uses the same Sobolev spaces, displayed gradients, target geometry, comparison class and energy normalization. In particular, small-energy regularity is a proved input rather than a new axiom or a field added to a minimizer structure. The dependency structure is shown in Figure 2.

4.1. **The joint Luckhaus construction.** Luckhaus interpolation connects two Sobolev maps on a sphere by an ambient-valued map on a thin cylinder. Its energy and distance estimates must be obtained for the same connector. The distance estimate allows tubular projection. The energy estimate then controls the projected competitor. Separate existential connectors would not support this argument.

For $A = \int_{S^{n-1}} (|\nabla_S u|^2 + |\nabla_S v|^2)$ and $D = \int_{S^{n-1}} |u - v|^2$, the joint estimates are

$$\int_{S^{n-1}\times(0,\epsilon)} |\nabla w|^2 \le C_n \epsilon A + C_n \epsilon^{-1} D, \tag{4.1}$$

$$\operatorname{dist}(w, N)^2 \le C_n \epsilon^{1-n} \sqrt{A}\sqrt{D} + C_n \epsilon^{-n} D \quad \text{a.e.} \tag{4.2}$$

The theorem uses $0 < \epsilon < 1/8$. Its exact quantifiers and trace content are in Proposition A.8. This range is substantive. An unrestricted all-thickness version of the distance estimate is false for general compact targets. For $N = \{0, 1\} \subset \mathbb{R}$, constant endpoints $u = 0, v = 1$ have $A = 0$, while a Sobolev connector crosses the gap. The right side of (4.2) tends to zero as $\epsilon \to \infty$. The stated small-cylinder theorem is what the compactness argument needs.

The construction proceeds on a translated cubical grid. Interpolation on the one-skeleton is followed by conical extension on higher-dimensional face cylinders. A suitable cube boundary is transferred to the sphere. In Lean, each step carries the map, weak-gradient columns, integrability, endpoint traces, common-face compatibility, energy bounds and distance bounds. The map on the final cylinder also has a finite annular Sobolev extension with the required radial coherence.

The core source files are `OriginalUniformPrismFamily.lean`, `CanonicalPrismSuccessor.lean`, `LuckhausJointHighDimensional.lean` and `ProjectedLuckhausAnnularConnector.lean`. The scalar estimates are organized separately in the weighted energy recurrence and tube recurrence.

4.2. **Compactness and actual tangent limits.** The projected connector joins an interior comparison map to an exterior minimizer. Trace and zero-boundary results make it an admissible ball competitor. The comparison proves that weak limits remain minimizing and that the local energies converge. Weak convergence of the gradients, together with convergence of their squared Hilbert norms, then yields strong convergence. Proposition A.9 records the subsequence result.

There are two logically distinct uses of compactness. Extraction gives a subsequence and a limit. If the original sequence already has a prescribed weak map limit, every convergent subsequence must have that same limit. The formalization uses this identification to obtain strong convergence of the

whole original sequence and the moving-center density inequality in Proposition A.10. Assuming limit minimality or energy convergence in that statement would omit the main compactness argument.

Applying the local compactness theorem on expanding rescaled domains gives global tangent maps along any prescribed sequence of shrinking positive scales. Monotonicity implies the same energy at every radius for the limit, and its radial derivative vanishes. Hence the tangent is homogeneous. The raw tangent relation is defined by actual strong local convergence. Target values, minimality, homogeneity and the energy identity are proved for every raw tangent, rather than inserted into that relation as assumptions.

4.3. **Spines, concentration and covering.** For a homogeneous minimizing tangent, the maximal-density set is a real linear subspace. Translation by a vector in this subspace leaves the map unchanged almost everywhere. This gives the spine in Proposition A.12. Density upper semicontinuity then forces near-maximal-density points at small scales to lie close to a plane whose dimension is controlled by the tangent spine.

The concentration statement is local and qualitative. It is converted into Hausdorff bounds by splitting the stratum into suitable density pieces and applying iterated affine covering estimates. The proof does not establish quantitative stratification or rectifiability.

The exclusion of spines of dimension greater than $n-3$ identifies the top three strata with the singular set. In dimension three, a homogeneous minimizing tangent is regular away from the origin. A nonzero singular point would generate a singular ray segment, contradicting the preliminary $\mathcal{H}^1$-nullity of its singular set. If distinct singular points accumulated in the original domain, rescaling by their distances and passing to a subsequence would give a nonzero limiting direction. Moving-center density upper semicontinuity and the uniform singular density gap would force positive density there, contradicting that regularity. The separate two-dimensional endpoint follows from local absolute continuity of the energy and small-energy regularity.

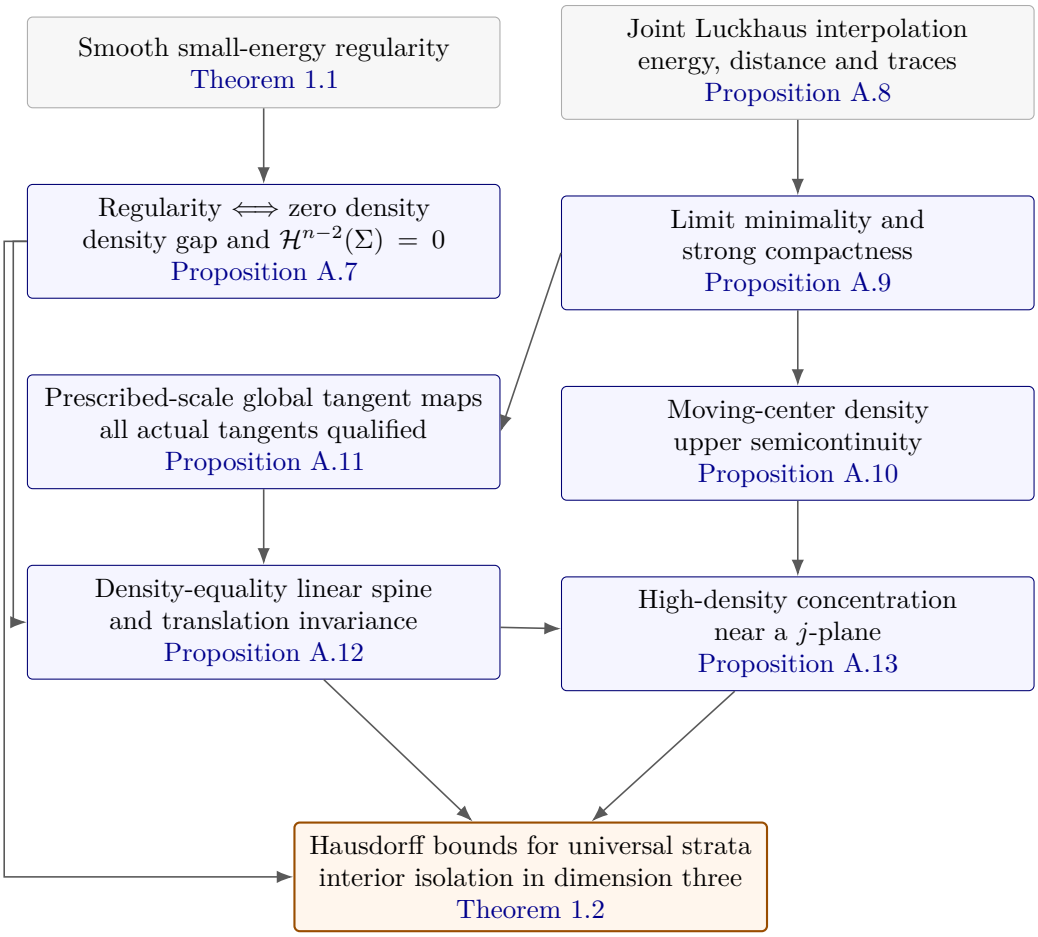


FIGURE 2. The stratification argument, with links to the actual statements. Monotonicity and affine transport are shared background inputs to tangent extraction. The arrow from regularity to the final theorem includes the density gap and preliminary Hausdorff nullity needed for the spine bound and interior isolation.

5. Workflow and mathematical supervision

5.1. **The supplied mathematics.** The workflow began with a mathematical goal and a detailed natural-language proof. For regularity, I separated small energy to Hölder continuity from Hölder continuity to smoothness. Each task received its own proof description and interfaces. After the two parts were completed, Codex was asked to connect them. The stratification task was then given the Giaquinta–Martinazzi route and instructed to import the established regularity library.

The agent used the installed Lean skills developed by Scott Armstrong, at revision `05a073 11f16bc581729e52f3e73647b95fa9bef1`. The installation manifests list statement auditing, dependency-graph construction, orchestration, elaboration, proof patterns and verification skills. Invoking these skills required ordinary prompts referring to the installed collection. There was no project-specific model training. This should not be confused with a one-prompt experiment. The retained records contain continuation requests, requests for multi-agent work, statement checks, integration requests and mathematical steering.

The skills helped keep the mathematical statement separate from the proof implementation. The public interface fixes the assumptions, quantifiers, normalization and shared witnesses. Proof workers implement dependencies, while separate reviews check semantic fidelity and the final dependency closures. Compiling a theorem is insufficient if its assumptions have absorbed the theorem one intended to prove.

The principal recorded model configuration was `gpt-6.1-sol` with reasoning effort `xhigh`. The retained session metadata also records some use of other model configurations. This is a description of the development environment, rather than a controlled comparison of models or tools.

5.2. **The higher-dimensional induction.** At the start of the stratification task, Luckhaus interpolation was the expected bottleneck. Its construction combines subdivisions, Sobolev slicing, compatible representatives, traces, conical extensions and projection. These operations are geometrically natural on paper, but their compatibility has to be expressed precisely in Lean.

During the construction, progress messages suggested that the agent was continuing with a specific four-dimensional case. I used a side chat to inspect the direction and redirected the main task toward a general inductive construction. A retained instruction on October 16 explicitly asks the agent to avoid expanding dimensions four, five and six one by one, and instead to preserve shared traces, energy and distance bounds in a uniform induction state. The side-chat inspection is my account of the interaction. The main-task instruction and the final generic code provide independent evidence of the redirection and its implementation.

The resulting `CoherentTwoStepRecursion.lean` uses genuine recursion. Its application in `OriginalUniformPrismFamily.lean` combines initial ranks with a general analytic successor. The state retains the lower-rank choices and their actual traces, gradients and budgets. Some fixed-dimensional components remain in the repository. The mathematical point is that the general result is obtained by a uniform induction rather than by a list of dimension-specific theorems. The intervention avoided further repetition, although the time saved cannot be measured from the available records.

5.3. **Why gluing was expensive.** The published stratification repository contains 712 Lean modules and 78,705 lines, counting blank lines and comments. The project-level import closure of the joint Luckhaus declaration contains 373 modules and 45,920 lines. This includes shared infrastructure, so it is not an exclusive measure of code devoted to one lemma. It nevertheless makes the size of its dependency burden visible. The interpolation directory alone contains 90 modules.

The difficulty was not primarily the last scalar inequality. A piecewise map requires a carrier, a measurable representative, a weak derivative, matching traces and proof that no boundary term remains. The recurrence for its energy can often be checked with much shorter algebra once these objects exist. This suggests a useful division for larger projects: separate the construction and

gluing of functions from the numerical estimates they satisfy, but keep the two connected by a joint theorem about the same witnesses.

6. Verification, duration and limitations

Both repositories pin Lean 4.32.2 and Mathlib revision `905b95818eb32af7874a58b427f50c1711a5e96c`. The stratification repository pins the regularity dependency at `f3a3097025ab9ddd956859600211d1d4edf0654e`. The repository citations and source links in this article refer to fixed revisions, rather than moving branches.

The regularity development contains 376 project modules. Its statement checks protect 15 complete theorem interfaces. The stratification development contains 712 project modules, 42 protected interfaces and 493 registered public exports. Source, statement and registered-closure axiom checks were rerun for this article and passed. The permitted axioms are `propext`, `Classical.choice` and `Quot.sound`. The repository verification records also report successful builds and kernel replay. The scripts and the replay have their documented coverage. An axiom check is not a substitute for checking that the declared type expresses the intended mathematics.

The source predicates use actual distributional derivatives and local L^2 data. Zero-boundary values are defined by closure of smooth compactly supported maps in the function-and-gradient norm. Compact smooth target geometry is supplied as geometry, while tubular projection and regularity are proved consequences. These choices matter to the interpretation of the endpoints.

My elapsed-time account is that the formalization campaign took less than one week. The retained materials are consistent with work beginning around October 10 and publication of the stratification repository at 02:36 on October 17, Beijing time. The directly inspected user-session records begin on October 12, and the public release histories do not record the whole development. Thus the precise sub-week duration is an author-reported calendar estimate, not an independently reconstructed measurement of all work. It excludes earlier mathematical preparation, the development of the skills and the libraries reused here. No total human-hour, compute-hour or token-efficiency claim is made.

The experience supports a limited but useful conclusion. Given an appropriate proof, substantial existing analysis infrastructure, installed Lean skills and mathematical supervision, an agent can complete a sizeable formalization over a short development period. It does not show that all such projects are easy or that supervision is unnecessary. Codex and tools such as Claude Code make this workflow accessible, but this project does not compare their performance.

Code quality remains a separate question. Kernel checking certifies the proof terms against their types. It does not certify a good API, minimal duplication, fast builds, readable proofs or ease of maintenance. Meaningful follow-up criteria include the size of public dependency closures, reuse of generic lemmas, compilation cost and independent review of the mathematical interfaces. Proof selection and dimension-independent induction can improve these criteria before any local code cleanup.

Appendix A. Statements behind the diagrams

The assumptions of Section 1 remain in force for harmonic-map results. The statements below record the mathematical content of the named public declarations. Analytic results concerning a general coefficient A have their assumptions stated separately. Internal implementation details are linked at the declarations, including the explicit weak-gradient and trace objects used in Lean.

Proposition A.1 (Projection and Euler equation)**.** *There is a tubular neighborhood of N with a smooth nearest-point projection π satisfying $\pi|_N = \mathrm{id}$ and $D\pi(p)$ equal to orthogonal projection onto T_pN for $p \in N$. On a smaller uniform neighborhood its first two derivatives are bounded. There is a smooth ambient bilinear coefficient A whose restriction to N gives, for every minimizing map, the*

signed distributional equation

$$\Delta u = \sum_i A(u)[D_i u, D_i u].$$

Equivalently, for compact smooth tests ψ, $\int Du : D\psi = -\int \sum_i A(u)[D_i u, D_i u] \cdot \psi$.

The projection theorem, outer-variation theorem and representative equation give these interfaces.

Proposition A.2 (Monotonicity)**.** *For every smooth compactly supported vector field* X *on* Ω,

$$\int_\Omega \left(|Du|^2_{\mathrm{HS}} \operatorname{div} X - 2 \sum_{i,j} \langle D_i u, D_j u \rangle \, \partial_i X^j \right) \mathrm{d}x = 0.$$

For $0 < r < R$ *and* $\overline{B_R(a)} \subset \Omega$,

$$\Theta_u(a, R) - \Theta_u(a, r) = 2 \int_{B_R(a) \setminus B_r(a)} |x - a|^{2-n} |\partial_{\mathrm{rad},a} u|^2 \, \mathrm{d}x \geq 0.$$

The density $\Theta_u(a)$ *is finite and nonnegative and equals the infimum of* $\Theta_u(a, r)$ *over admissible radii.*

The regularity library proves inner stationarity and then monotonicity. The density characterization is exported in `DensityCharacterization.lean`.

Lemma A.3 (One-step decay)**.** *For each* n, N *there are* $\varepsilon_{\mathrm{dec}} > 0$ *and* $0 < \tau < 1/8$ *such that a minimizing map on the unit-ball comparison domain with* $E(u; B_1) \leq \varepsilon_{\mathrm{dec}}$ *satisfies*

$$\Theta_u(0, \tau) \leq \tfrac{1}{2} E(u; B_1).$$

The constants are uniform over the maps and displayed weak gradients.

See `HarmonicMaps.oneStepEnergyDecay`.

Proposition A.4 (Quantitative Hölder representative)**.** *For each* n, N *there exist* $\varepsilon_H > 0$, $0 < \alpha < 1$ *and* $C_H \geq 0$ *such that, if* $\Theta_u(a, R) \leq \varepsilon_H$ *on an admissible ball,* u *has a continuous representative* w *on* $B_{R/2}(a)$*, with values in* N*, and*

$$|w(x) - w(y)| \leq C_H R^{-\alpha} \sqrt{\Theta_u(a, R)} \, |x - y|^\alpha \qquad (x, y \in B_{R/2}(a)).$$

Any other continuous representative agrees with w *on that half ball.*

See `HarmonicMaps.smallEnergyRegularity`.

Proposition A.5 (Quadratic-Poisson C^1 gate)**.** *Fix* $n \geq 2$, q, $0 < \alpha < 1$ *and* $K \geq 0$*. There exist* $H_* > 0$ *and* $C_G \geq 0$*, depending only on these inputs, with the following property. Let* $w \in W^{1,2}(B_1, \mathbb{R}^q)$ *be continuous, with displayed weak gradient* Dw*, and let* $\Delta w = F$ *weakly, where* $|F| \leq K |Dw|^2_{\mathrm{HS}}$ *almost everywhere. If* $[w]_\alpha \leq H_*$*, then* w *is* C^1 *on* $B_{1/2}$ *and, with* $\gamma = \alpha/8$,

$$\sup_{B_{1/2}} \|Dw\| + [Dw]_{\gamma; B_{1/2}} \leq C_G \left(\int_{B_1} |Dw|^2_{\mathrm{HS}} \right)^{1/2}.$$

The derivative on the left is the classical derivative of w *and agrees almost everywhere with its displayed weak gradient. Bounds expressed separately with a common constant give this displayed sum after enlarging that constant if necessary.*

The frozen C^1 declaration records the precise ball and norm interfaces and the separate estimates.

Proposition A.6 (Weak Poisson Schauder estimate)**.** *Fix* $n \geq 2$, q *and* $0 < \gamma < 1$*. There is* $C_S \geq 0$*, depending only on* n, q, γ*, such that a continuous locally Sobolev map* v *satisfying* $\Delta v = f$ *weakly on* $B_R(a)$*, with* $f \in C^{0,\gamma}(B_R(a))$*, is* C^2 *on* $B_{R/2}(a)$ *and satisfies the classical equation there. For every constant vector* b,

$$R^2 \|D^2 v\|_{\infty; B_{R/2}} + R^{2+\gamma} [D^2 v]_{\gamma; B_{R/2}} \leq C_S (\|v - b\|_{\infty; B_R} + R^2 \|f\|_{\infty; B_R} + R^{2+\gamma} [f]_{\gamma; B_R}).$$

The statement applies when the displayed quantities on the right are finite. A common constant for the separate bounds can be enlarged to give the sum shown here.

See `PoissonInteriorSchauder.lean` and the proved arbitrary-ball rescaling.

Proposition A.7 (Regular points and density)**.** *For each n, N there is one $\varepsilon_0 > 0$ such that, for every minimizing map and every $a \in \Omega$,*

$$a \in \mathrm{Reg}(u) \iff \Theta_u(a) = 0, \qquad a \in \Sigma(u) \implies \Theta_u(a) \geq \varepsilon_0.$$

Moreover, $\Sigma(u)$ is relatively closed and $\mathcal{H}^{n-2}(\Sigma(u)) = 0$.

See `regularPoint_density_characterization` and `singularSet_hausdorffMeasure_zero`.

Proposition A.8 (Joint Luckhaus interpolation)**.** *For every $n \geq 2$ there is $C_n > 0$ chosen before the target dimension q, the compact target set $N \subset \mathbb{R}^q$, the sphere Sobolev maps $u, v \in W^{1,2}(S^{n-1}, N)$ and the thickness $0 < \epsilon < 1/8$, such that one ambient-valued Sobolev connector w has endpoint traces u, v and satisfies (4.1)–(4.2). The same construction has square-integrable native cylinder gradient columns, tangent spatial columns and a finite annular Sobolev extension with radial map coherence and almost-everywhere weak-column coherence. All trace, energy and distance conclusions concern the same witnesses. The connector need not take values in N.*

See `StratificationTheory.luckhaus_joint`. Unlike the harmonic-map theorems, this interpolation result only needs a compact target set.

Proposition A.9 (Strong compactness with local energies)**.** *Let (u_i) be minimizing maps on a fixed open domain, with locally uniform energy bounds. There exist one strictly increasing subsequence map $\kappa : \mathbb{N} \to \mathbb{N}$ and one minimizing limit v such that*

$$u_{\kappa(i)} \to v \quad \text{strongly in } W^{1,2}_{\mathrm{loc}}(\Omega).$$

The maps and all displayed weak-gradient columns converge strongly in L^2 on every compact subset. The actual Hilbert–Schmidt energy integrals and extended Dirichlet energies converge on compact subsets and admissible open balls, and the difference-gradient energy tends to zero there, for these same witnesses.

The original `strongCompactness` uses finite $W^{1,2}$ inputs on the domain. The local-input compactness theorem used for tangent extraction has the local hypotheses stated here. Its energy consequences follow from the strong-gradient energy lemmas on the same witnesses.

Proposition A.10 (Prescribed weak limit and density)**.** *Suppose (u_i) is a locally energy-bounded minimizing sequence and has a prescribed weak local L^2 map limit v. Then v has a displayed weak gradient, is locally Sobolev, N-valued and minimizing. The whole original sequence converges strongly locally in maps and gradient columns, with local ball energy convergence. For every $y_i \to y \in \Omega$,*

$$\limsup_i \Theta_{u_i}(y_i) \leq \Theta_v(y).$$

Weak local convergence is tested against every L^2 function on every compact subset. See `exists_prescribed_weak_limit_density_joint_upper_bound`.

Proposition A.11 (Tangents along prescribed scales)**.** *For every minimizing u, $a \in \Omega$ and prescribed positive sequence $\rho_i \to 0$, there exist one strictly increasing κ and one global minimizing map $\varphi \in W^{1,2}_{\mathrm{loc}}(\mathbb{R}^n, N)$ such that $u(a + \rho_{\kappa(i)} \cdot)$ and its rescaled weak-gradient columns converge strongly on every compact subset to $\varphi, D\varphi$. For all $R > 0$ and $\lambda > 0$,*

$$R^{2-n} E(\varphi; B_R) = \Theta_u(a), \qquad \varphi(\lambda \cdot) = \varphi \quad \text{a.e.}$$

Every raw tangent limit, defined by actual strong local map-and-gradient convergence along shrinking scales, has these qualifications. Uniqueness of tangents is not asserted.

See the prescribed-scale extraction, the raw tangent relation and its qualification theorem.

Proposition A.12 (Spine characterization)**.** *For a homogeneous global minimizing map φ, there is a unique real linear subspace $S(\varphi) \subset \mathbb{R}^n$ whose carrier is*

$$\{z : \Theta_\varphi(z) = \Theta_\varphi(0)\}.$$

For every z, $\Theta_\varphi(z) \le \Theta_\varphi(0)$, and for every $z \in S(\varphi)$, $\varphi(\cdot + z) = \varphi$ almost everywhere. At a singular point of a minimizing map, every raw tangent is nonconstant and, for $n \ge 3$, has spine dimension at most $n - 3$.

The density-equality characterization is `spine_characterization`. The singular-tangent dimension restriction is used in the complete stratification theorem.

Proposition A.13 (Concentration near a plane)**.** *Let $n \ge 3$, $0 \le j \le n$, $a \in S_j(u)$ and $\delta > 0$. There exists $\eta > 0$ such that for every $0 < \rho \le \eta$ with $\overline{B_\rho(a)} \subset \Omega$, there is a j-dimensional linear subspace $L \subset \mathbb{R}^n$ satisfying*

$$x \in B_\rho(a), \quad \Theta_u(x) \ge \Theta_u(a) - \eta \quad \Longrightarrow \quad \operatorname{dist}((x-a)/\rho, L) < \delta.$$

The plane may depend on ρ. The same η controls the scale and the density tolerance.

This is a clause of `stratification`. The actual affine approximation and Hausdorff covering interface is in `DeltaApproximationCover.lean`.

Appendix B. Alternative regularity routes

B.1. **Direct smoothing and comparison.** The route used here is the Schoen–Uhlenbeck argument as presented in Giaquinta–Martinazzi, Section 10.3.1. Starting with a small energy bound $E(u; B_1) \le \eta$, the formal comparison chooses smoothing and geometric scales of the form

$$\bar{h} = \eta^{1/4}, \qquad \delta = \eta^{1/8}, \qquad \tau = \eta^{1/(64(n-1))}.$$

The choices in this construction are made with a small upper energy parameter fixed uniformly over the maps. Monotonicity controls the smoothed gradients at every scale. A projected harmonic replacement reduces the interior energy, and a projected variable smoothing restores the original boundary data on a shell. Minimality converts the competitor bound into the contraction in Lemma A.3.

The formal library proves the smoothing, projection and admissibility facts used by this comparison. The route makes a quantitative dependency chain available for inspection. Turning it into numerical data still requires numerical geometric bounds and numerical versions of every analytic constant, as explained in Section 3.

B.2. **Reverse Poincaré and harmonic approximation.** Simon [8] gives a closely related regularity framework based on a Caccioppoli, or reverse Poincaré, estimate and harmonic approximation. These are usually two parts of one proof architecture, rather than two unrelated proofs. In a typical normalized form, comparison gives

$$r^{2-n} \int_{B_r} |Du|^2 \le C \left(r^{-n} \int_{B_{2r}} |u - u_{B_{2r}}|^2 + \text{controlled comparison error} \right).$$

The precise error and admissible radii depend on the chosen version. Luckhaus interpolation supplies competitors when direct target-valued cutoff interpolation is unavailable.

Harmonic approximation then compares a normalized small-energy map with a harmonic map. In the customary qualitative proof, a failure sequence has a weakly convergent subsequence. The rescaled nonlinear equation becomes linear in the limit, while compactness and the comparison estimate give the convergence needed for decay. Harmonic interior estimates improve the energy on a smaller ball. Iteration gives Hölder continuity, after which the elliptic upgrade applies.

Qualitative harmonic approximation often asserts that for each desired error there is a sufficiently small defect. It does not supply a numerical modulus. This is an additional extraction problem. Harmonic approximation itself is not inherently nonquantitative: a proof with an explicit modulus could serve the same quantitative purpose.

B.3. **Compactness and contradiction.** The compactness viewpoint in minimizing-map regularity is associated with Schoen–Uhlenbeck and Hardt–Lin, and its robust interpolation-based form uses Luckhaus' theorem [7, 5, 6, 4]. A decay argument begins by assuming that no uniform threshold works. Choose a sequence of counterexamples with energy tending to zero, subtract suitable means and normalize their energy. Compactness gives a subsequence. The limiting map satisfies a linear harmonic equation. Strong energy convergence transfers the harmonic decay estimate to the counterexamples and yields a contradiction.

The occurrence of a subsequence is logically sufficient. If every proposed threshold fails, one can choose an entire counterexample sequence, and a contradictory subsequence already rules out that failure. It does not weaken the final theorem to a statement about a subsequence of source maps. What the usual argument withholds is an explicit threshold as a function of the desired decay and the geometric data.

For this project, the direct route gave regularity before the extensive gluing machinery was available. Compactness was then formalized where it was indispensable for tangent maps and stratification. This order is a practical choice of dependencies, rather than a claim that one classical proof is mathematically stronger.

Appendix C. Details of the quantitative extraction

All formulas in this appendix refer to the same energy and derivative norm conventions as the main theorem. Local arithmetic witnesses are visible in the linked Lean proofs. Closed-form evaluations, explicit replacements for limit-selected scales, and the conservative kernel bounds below are additional mathematical deductions. They have not been added as new Lean lemmas.

C.1. **From one-step decay to the initial Hölder bound.** The proof in `MorreyHolder.lean` uses

$$\mathcal{M}_{n,\alpha} = 8\left[\frac{2\,2^{\alpha}\sqrt{2^{2n+1}/v_n}}{1-2^{-\alpha}} + 2\,3^{\alpha}\sqrt{2^{n+1}(3/2)^n/v_n}\right], \qquad C_H = \mathcal{M}_{n,\alpha}\sqrt{2^{n-1+2\alpha}}. \tag{C.1}$$

The bridge to the enlarged-ball representative in `LocalHolder.lean` chooses

$$P = 4^{n-2}, \qquad \varepsilon_{\mathrm{enl}} = \varepsilon_H/P, \qquad L = (C_H 4^{\alpha}\sqrt{P}+1)24^{1-\alpha}. \tag{C.2}$$

Thus the representative on $B_{3/4}$ has Hölder coefficient at most $L\sqrt{E}$.

C.2. **The C^1 gate.** Let κ_1 bound the first derivative of the fixed normalized mollifier. The harmonic comparison proof exhibits

$$c = 2^{n+1} + 12C_{\mathrm{osc}} + 5, \qquad C_{\mathrm{osc}} = \kappa_1^2\,2^{2n+2}v_n^2.$$

Set $r_0 = 1/16$, $\gamma = \alpha/8$, $\lambda = n - \alpha/2$, $\nu = n + \alpha/4$ and $\mu = n + \alpha/2$. Admissible explicit scales and the associated constants are

$$\begin{aligned}
&\theta_E = \min\{1/4, (4c)^{-2/\alpha}\}, && \theta_\Phi = \min\{1/4, (4c)^{-1/(2-\alpha/4)}\},\\
&H_* = \min\left\{1, \frac{\theta_E^{\lambda}}{4c(1+K)r_0^{\alpha}}\right\}, && C_E = (\theta_E r_0)^{-\lambda},\\
&C_\Phi = \theta_\Phi^{-\nu}\left(r_0^{-\nu} + 2cKC_E\theta_\Phi^{-\nu}r_0^{\mu-\nu}\right).
\end{aligned} \tag{C.3}$$

The choices satisfy the scalar absorption inequalities in `ImprovedGrowthParameters.lean`. The source presently obtains its small scales from convergence of powers. The bounds for energy

growth and gradient excess are the displayed witnesses in `ImprovedEnergyGrowth.lean` and `GradientExcessDecay.lean`.

The Campanato constants can be written as

$$\begin{aligned} J &= \sqrt{2^n/v_n}, & T &= J/(1-2^{-\gamma}),\\ U &= (r_0^n v_n)^{-1/2} + T\sqrt{C_\Phi}\, r_0^\gamma, & V &= (2T2^\gamma + 2J4^\gamma)\sqrt{C_\Phi},\\ G &= U + V + 2U/(r_0/4)^\gamma + 1, & C_G &= \sqrt{n}\, G. \end{aligned}$$

These follow the named constants in `Campanato/Representative.lean` and the gradient reconstruction. The initial all-order bound and common smallness threshold are

$$M_1 = L + C_G(1 + 2^{1-\gamma}) + 1, \qquad \varepsilon_* = \min\{1, \varepsilon_{\mathrm{enl}}, (H_*/L)^2\}. \tag{C.4}$$

The important feature is that (C.4) is chosen before any derivative order.

C.3. **Unit and nested Schauder coefficients.** For derivative bounds κ_j of the normalized kernel, $0 \le j \le 3$, and a bound a_1 for $|s'|$, set

$$\begin{aligned} C_F &= 8a_1 + 9, & C_{\mathrm{harm}} &= \sum_{j=0}^{3} 8^j 2^{n+j} v_n \kappa_j,\\ V_2 &= \tfrac{9}{2}\log 3 - \tfrac{7}{4}, & V_n &= \frac{9}{2(n-2)} \quad (n>2),\\ P_H &= \frac{1}{n} + \frac{(n+1)3^n}{n} + \frac{n+1}{\gamma}, & C_P &= V_n + 3\sqrt{n} + nP_H,\\ J_H &= \frac{1}{n} + \frac{(n+1)(2^\gamma + 3^\gamma)}{\gamma} + \frac{n(n+5)2^{n+\gamma}}{1-\gamma} + (n+1)\log 3. \end{aligned}$$

An admissible unit Schauder coefficient for the separate derivative and Hölder estimates is

$$C_S = 1 + C_{\mathrm{harm}}(1 + V_n) + (C_P + nJ_H)(1 + C_F). \tag{C.5}$$

This is the aggregation witness in the Schauder provider, after elementary evaluation of the potential integrals. For $\rho_m = 2^{-m-4}$ and $\Delta_m = 2^{-m-3}$, put

$$S_m = C_S(1 + \rho_m^2 + \rho_m^{2+\gamma})\left[(1 + \rho_m^{-1} + \rho_m^{-2})\{1 + 2(\Delta_m/8)^{-\gamma}\} + \rho_m^{-2-\gamma}\right]. \tag{C.6}$$

The radius choices and local-to-global Hölder estimate occur in `InteriorLocalization.lean`. These are separate-bound coefficients. If the two bounds are written as one sum, their sum is bounded using $2C_S$.

C.4. **Forcing and composition recurrences.** Let $b_i \ge \sup \|D^i A\|$ on the fixed closed range ball. Convexity gives b_{i+1} as a Lipschitz bound for $D^i A$. For $B, T, L \ge 0$ define

$$\begin{aligned} \mathcal{H}_0(B,T,L) &= BL,\\ \mathcal{H}_{j+1}(B,T,L) &= \mathcal{H}_j(B,T,L) + 2^j T\{\mathcal{H}_j(B,T,L) + j!B\max(1,T)^j\},\\ P_j(n,B,T) &= n4^j j!B\max(1,T)^j T^2,\\ Q_j(n,B,T,L) &= n4^j T^2\{\mathcal{H}_j(B,T,L) + 2j!B\max(1,T)^j\}. \end{aligned} \tag{C.7}$$

At step m, use $B = \sum_{i=0}^m b_i$, $T = M_m \max\{1, (2r_m)^{1-\gamma}\}$ and $L = M_m(2r_m)^{1-\gamma}$. Insert these arguments with $j = m-1$ into (3.4). The formulas are the bodies of the composition and quadratic-forcing constants in `QuadraticComposition.lean`. The two differentiated positive-order factors retain the small parameter δ^2, which is bounded by δ because $0 \le \delta \le 1$.

C.5. **Eliminating the fixed-kernel choices.** Here are deliberately loose numerical bounds obtained directly from the fixed transition function. For $z = 1/t > 0$,

$$p' = z^2 e^{-z}, \quad p'' = (z^4 - 2z^3)e^{-z}, \quad p^{(3)} = (z^6 - 6z^5 + 6z^4)e^{-z}.$$

The inequality $z^d e^{-z} \le (d/e)^d$, together with $8/3 \le e < 3$, gives bounds $(F_0, F_1, F_2, F_3) = (1, 1, 8, 300)$. On $(0, 1)$, $p(t) + p(1-t) \ge e^{-2} > 1/9$. Differentiating $(p(t) + p(1-t))s(t) = p(t)$ yields the sufficient recursion

$$a_0 = 1, \qquad a_k = 9\left(F_k + 2\sum_{i=1}^{k} \binom{k}{i} F_i a_{k-i}\right),$$

so one may take

$$(a_0, a_1, a_2, a_3) = (1, 27, 1188, 83916).$$

Outside $[0, 1]$ the transition is constant, and smoothness handles the endpoints. Monotonicity gives $I_n \ge v_n 2^{-1-3n/2}$ by integrating over $B_{1/(2\sqrt{2})}$. The first three chain rules for $s(1 - 4|x|^2)$ therefore allow the choice

$$\kappa_j = \frac{2^{1+3n/2}}{v_n} R_j, \qquad (R_0, R_1, R_2, R_3) = (1, 108, 19224, 5484672).$$

In particular the extracted constants may use

$$C_F \le 225, \qquad C_{\mathrm{harm}} \le 22470139585\, 2^{1+5n/2}, \qquad c \le 2^{n+1} + 139968\, 2^{5n+4} + 5. \tag{C.8}$$

For example the last expression is 73383542805 when $n = 3$. At $\gamma = 1/64$, (C.5) with these upper bounds is approximately 6.03099×10^{12} in dimension two and 4.47428×10^{13} in dimension three. These are numerical evaluations of explicitly stated universal expressions, separate from the choice of a target.

C.6. **Sphere geometry and the scope of the numerical example.** For $|p| = 1$ the full ambient Hessian of nearest projection is

$$D^2\pi(p)[v, w] = -p\langle v, w\rangle - v\langle p, w\rangle - w\langle p, v\rangle + 3p\langle p, v\rangle\langle p, w\rangle.$$

Its norm is at most $6\|v\|\|w\|$. On the $1/4$-tube, $\|D\pi\| \le 4/3$. For a global extension choose $p_0 \in N$ and

$$\chi(r) = s(4r - 2)\{1 - s(4r - 5)\}, \qquad Q(z) = \chi(|z|)\frac{z}{|z|} + (1 - \chi(|z|))p_0.$$

Set $Q(0) = p_0$. The cutoff equals one on $[3/4, 5/4]$ and vanishes for $r \le 1/2$ and $r \ge 3/2$. Thus Q is smooth, agrees with π on the whole tube, and has bounded global derivatives. The $4/3$ derivative bound is asserted only on the tube.

The Sobolev chain rule for $|u|^2 = 1$ gives $\langle u, D_i u\rangle = 0$ almost everywhere. The sphere equation therefore uses the alternative smooth coefficient $A(p)[v, w] = -p\langle v, w\rangle$. It is affine, and its derivatives on $\overline{B}_1$ have bounds $b_0 = b_1 = 1$ and $b_j = 0$ for $j \ge 2$. It is not globally bounded on $\mathbb{R}^q$ and is not the full Hessian of Q. The generic bootstrap only uses its bounds on the range ball. Combining these geometric data with (3.2)–(3.3) and (C.1)–(C.8) gives the specialization in Section 3. This recombination and sphere specialization have not been formalized as a new Lean export.

The standalone evaluator uses Python's decimal arithmetic at 70-digit working precision. It is useful for checking the displayed arithmetic. It does not certify the validity of caller-supplied geometric bounds or provide directed-rounding interval certificates. The exact mathematical expressions, rather than the rounded decimal strings, are the candidate upper bounds.

Appendix D. Reproduction and source correspondence

The public endpoints can be imported as follows:

```
module
public import HarmonicMaps.Regularity
#check HarmonicMaps.smallEnergyRegularity
#check HolderToCK.smallEnergySmoothRegularity
```

In the stratification project:

```
module
public import StratificationTheory
#check StratificationTheory.luckhaus_joint
#check StratificationTheory.stratification
#check StratificationTheory.exists_tangent_along_prescribed_scales
```

The reproducible commands, after fetching the pinned dependencies, are listed in the respective READMEs. They include `lake build`, source checks, statement checks and axiom checks. Kernel replay is a separate command and workflow. The regularity and stratification verification instructions and verification guide specify their exact coverage.

The mathematical input manuscripts are retained in each repository's `sources` directory. Some input text records proof obligations or source issues. It is not itself a certificate of the final proof status. The public declarations, their checked proof closures and the current mathematical guides determine what is claimed in this article.

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